unit 1 MCQ MTH165

Mathematics (Lovely Professional University)

MCQ PRACTICE
MTH 165

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Chapter 1

Matrix Algebra

1. The inverse of the matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

   (a) I (b) A
   (c) $-A$ (d) $2A$

2. The inverse of the matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

   (a) I (b) A
   (c) $-A$ (d) $2A$

3. The inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

   (a) $A^2 - 2I$ (b) $2I + A^2$
   (c) $A^2 + I$ (d) $A^2 - I$

4. The inverse of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$

   (a) $A^2 - 2I$ (b) $2I + A^2$
   (c) $A^2 + I$ (d) $A^2 - I$

5. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}$ then $2A^{-1} =?$

(a) A^2+3I
(b) $3I-A^2$
(c) A^2-3I
(d) A^2-I

6. If A is invertible symmetric matrix, then

(a) A^{-1} is symmetric
(b) A^{-1} is skew-symmetric
(c) $A^2-2A+I=0$
(d) None of These

7. If the matrix $A=\begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric then x is equal to

(a) 2
(b) 3
(c) 4
(d) 5

8. If $\begin{bmatrix} a+b & 3 \\ 5 & ab \end{bmatrix}=\begin{bmatrix} 4 & 3 \\ 5 & 3 \end{bmatrix}$ then what are values of a and b?

(a) $(2,4)$ or $(4,2)$
(b) $(0,3)$ or $(3,0)$
(c) $(2,1)$ or $(1,2)$
(d) $(1,3)$ or $(3,1)$

9. If the matrix $A=\begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix}$ and $B=\begin{bmatrix} 3 & 4 \\ 7 & 1 \end{bmatrix}$ then $4A-3B=$?

(a) $\begin{bmatrix} -5 & -20 \\ -9 & 17 \end{bmatrix}$
(b) $\begin{bmatrix} 5 & 20 \\ 9 & -17 \end{bmatrix}$
(c) $\begin{bmatrix} -2 & -2 \\ -4 & 4 \end{bmatrix}$
(d) $\begin{bmatrix} -5 & 20 \\ 9 & 17 \end{bmatrix}$

10. If the matrix $B=\begin{bmatrix} 1 & 7 \\ 3 & 1 \end{bmatrix}$ and $C=\begin{bmatrix} 4 & 1 \\ 6 & 8 \end{bmatrix}$ and $2A+3B-6C=O$ then what is value of A?

(a) $\begin{bmatrix} 21/2 & 27/2 \\ -15/2 & 45/2 \end{bmatrix}$
(b) $\begin{bmatrix} 21/4 & 27/4 \\ -15/4 & 45/4 \end{bmatrix}$
(c) $\begin{bmatrix} 21/4 & -15/4 \\ 27/4 & 45/4 \end{bmatrix}$
(d) $\begin{bmatrix} 21/2 & -15/2 \\ 27/2 & 45/2 \end{bmatrix}$

11. If the matrix $A=\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix such that $AA^T=9I$, then what are the values of a and b ?

(a) $a=-1, b=-2$
(b) $a=-2, b=-1$
(c) $a=1, b=2$
(d) $a=2, b=1$

12. If the matrix $A=\begin{bmatrix} x & 2 & 0 \\ 2 & 0 & 1 \\ 6 & 3 & 0 \end{bmatrix}$ is a singular matrix then what is the value of x?

(a) 0
(b) 2
(c) 4
(d) 6

13. If the matrix $A = \begin{bmatrix} k & 0 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 \\ 6 & 16 \end{bmatrix}$ then what is value of k for which $A^2 = B$?

(a) -1
(b) -2
(c) 1
(d) 2

14. Suppose that A is a square matrix, then $A + A^T$ is

(a) Not a square matrix
(b) a skew symmetric matrix
(c) a symmetric matrix
(d) None of these.

15. The Eigenvalues of $\begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$ is

(a) $1, 4$
(b) $2, 3$
(c) $0, 5$
(d) $1, 5$

16. The Eigenvalues of $\begin{bmatrix} 4 & 2 & 17 \\ 0 & -3 & 23 \\ 0 & 0 & 3 \end{bmatrix}$ is

(a) $-3, 4, 3$
(b) $2, -3, 5$
(c) $-2, 6, 3$
(d) $3, 3, 4$

17. The Eigenvalues of $\begin{bmatrix} 1 & 2 & -2 \\ 0 & 2 & 2 \\ 0 & 1 & 3 \end{bmatrix}$ is

(a) $-1, 4, 2$
(b) $4, -3, 5$
(c) $1, 2, 3$
(d) $1, 1, 4$

18. The Eigenvalues of $\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$ is

(a) $1, 4, 4$
(b) $1, 4, -4$
(c) $3, 3, 3$
(d) $1, 2, 6$

19. The Eigenvalues of $\begin{bmatrix} -2 & 2 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 0 \end{bmatrix}$ is

(a) $0, 2\sqrt{2}, -2\sqrt{2}$
(b) $0, -2, 2$
(c) $0, 1, 2$
(d) $0, \sqrt{2}, -\sqrt{2}$

20. The Eigenvalues of $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$ is

(a) $0, 0, \sqrt{6}$ (b) $0, 1, 5$

(c) $0, 0, 6$ (d) $0, 3, 3$

21. The Eigenvalues of $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 3 & 3 \end{bmatrix}$ is

(a) $1, 1, 4$ (b) $0, 1, 5$

(c) $0, 0, 6$ (d) $1, 2, 3$

22. If for the matrix $\begin{bmatrix} 3 & 0 & 1 \\ -1 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$ one of eigenvector is $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$. Find the eigenvalue corresponding to this vector.

(a) 1 (b) 2

(c) 3 (d) 4

23. If for the matrix $\begin{bmatrix} 3 & 0 & 1 \\ -1 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$ one of eigenvector is $\begin{bmatrix} -1/4 \\ -9/8 \\ 1 \end{bmatrix}$. Find the eigenvalue corresponding to this vector.

(a) -1 (b) 2

(c) 3 (d) 4

24. If for the matrix $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$ one of eigenvector is $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$. Find the eigenvalue corresponding to this vector.

(a) -1 (b) -3

(c) 4 (d) 3

25. If for the matrix $\begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 0 \\ 5 & -3 & 3 \end{bmatrix}$ one of eigenvector is $\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$. Find the eigenvalue corresponding to this vector.

(a) 1 (b) 2

(c) 3 (d) 4

26. If A is a real and symmetric, then the eigenvalues

(a) are always real
(b) are always real and positive
(c) are always real and non-negative
(d) occurs in complex conjugate pairs

27. Which of the following is **NOT** an eigenvector for the matrix $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$?

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$
(b) $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$
(c) $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$
(d) $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

28. A is a 3 x 4 real matrix and $Ax = b$ is an inconsistent system of equations. Then highest possible rank of A is

(a) 1
(b) 2
(c) 3
(d) 4

29. The characteristic equation of the matrix $\begin{bmatrix} 3 & 2 & 9 \\ 7 & 5 & 13 \\ 6 & 17 & 19 \end{bmatrix}$ is given by

(a) $\lambda^3 - 27\lambda^2 + 167\lambda - 285 = 0$
(b) $\lambda^3 - 27\lambda^2 - 122\lambda - 313 = 0$
(c) $\lambda^3 - 27\lambda^2 + 167\lambda + 285 = 0$
(d) $\lambda^3 - 23\lambda^2 + 167\lambda + 313 = 0$

30. The characteristic equation of the matrix $\begin{bmatrix} 2 & 3 & -1 \\ 3 & 2 & 0 \\ 5 & -3 & 3 \end{bmatrix}$ is given by

(a) $-\lambda^3 + 7\lambda^2 - 12\lambda + 4 = 0$
(b) $-\lambda^3 + 7\lambda^2 + 12\lambda + 4 = 0$
(c) $-\lambda^3 - 7\lambda^2 - 12\lambda + 4 = 0$
(d) $-\lambda^3 + 7\lambda^2 - 12\lambda - 4 = 0$

31. Two of the eigenvalues of a 3×3 matrix A are 2&3, and its determinant is 36 than the third eigenvalue is

(a) 2
(b) 3
(c) 4
(d) 6

32. The eigenvalues of 3×3 matrix A are given as $1, 2, 3$. The $\det(A)$ is given by

(a) -6
(b) 5
(c) 6
(d) None of these.

33. The eigenvalues of 4×4 matrix A are given as $4, -5, 3, 13$. The $\det(A)$ is given by

(a) -780
(b) 520
(c) 630
(d) None of these.

34. If one of the eigenvalues of A is zero, it implies

(a) The solution of linear equations $AX = 0$ is non-trivial.

(b) The determinant of A is non-zero.

(c) The solution to $AX = 0$ is unique.

(d) The matrix is non-singular.

35. Find the eigenvector for value of $\lambda = -2$ for the given matrix $A = \begin{bmatrix} 3 & 5 \\ 3 & 1 \end{bmatrix}$

(a) $A = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ (b) $A = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(c) $A = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ (d) $A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

36. If A is a 3×3 matrix having all entries as 2023, then rank of A is

(a) 3 (b) 2

(c) 1 (d) 0

37. For which values of k, the system of equations $x - 2y + z = 0$, $3x - y + 2z = 0$, $y + kz = 0$ have infinitely many solutions

(a) $k = \frac{1}{5}$ (b) $k \neq \frac{1}{5}$

(c) $k = -\frac{1}{5}$ (d) $k \neq -\frac{1}{5}$

38. Find x for which rank of the matrix $A = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & x \\ 1 & 2 & 3 \end{bmatrix}$ is less than 3.

(a) $k = 3$ (b) $k = -3$

(c) $k = 0$ (d) None of these

39. What is rank of the matrix $\begin{bmatrix} 3 & 0 & 1 \\ 1 & 2 & 4 \\ 1 & 2 & 3 \end{bmatrix}$

(a) 0 (b) 1

(c) 2 (d) 3

40. What is rank of the matrix $\begin{bmatrix} 2023 & 2023 & 2023 \\ 2023 & 2023 & 2023 \\ 2023 & 2023 & 2023 \end{bmatrix}$

(a) 0 (b) 1

(c) 2 (d) 3

41. What is rank of the matrix $\begin{bmatrix} 2024 & 2023 & 2023 \\ 2024 & 2023 & 2023 \\ 2024 & 2023 & 2023 \end{bmatrix}$

(a) 0
(b) 1
(c) 2
(d) 3

42. What is rank of the matrix $\begin{bmatrix} 3 & -1 & 2 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

(a) 0
(b) 1
(c) 2
(d) 3

43. What is rank of the matrix $\begin{bmatrix} 1 & 3 & 5 & 1 \\ 2 & 4 & 8 & 0 \\ 3 & 1 & 7 & 5 \end{bmatrix}$

(a) 1
(b) 2
(c) 3
(d) 4

44. What is rank of the identity matrix of order 4×4 ?

(a) 1
(b) 2
(c) 3
(d) 4

45. What is rank of the identity matrix of order 10×10 ?

(a) 100
(b) 10
(c) 1
(d) 0

46. For which value of k the following system of linear equations have unique solution:

$$\begin{aligned} 2x + 3y + 5z &= 9 \\ 7x + 3y - 2z &= 8 \\ 2x + 3y + kz &= 9 \end{aligned}$$

(a) $k = 15$
(b) $k = 5$
(c) $k \neq 15$
(d) $k \neq 5$

47. For which value of k the following system of linear equations have no solution:

$$\begin{aligned} 4x + 2y + z &= 0 \\ 3x - y + 3z &= -1 \\ x + ky + 2z &= 0 \end{aligned}$$

(a) $k = \frac{13}{9}$
(b) $k = \frac{9}{13}$
(c) $k = -\frac{13}{9}$
(d) $k = -\frac{9}{13}$

48. For which value of k the following system of linear equations have no solution:

$$\begin{aligned} 4x + 2y + z &= 3 \\ 3x - y + 3z &= -1 \\ x + y + kz &= -2 \end{aligned}$$

(a) $k = \frac{1}{5}$
(b) $k = -\frac{1}{5}$
(c) $k = -5$
(d) $k = 5$

49. Which of the following vectors in $\mathbb{R}^3$ are linearly independent ?

(a) $\{(1,2,1),(2,1,1),(1,1,2)\}$
(b) $\{(\frac{1}{2},3),(2,12)\}$
(c) $\{(2,2,2),(2,1,1),(1,1,1)\}$
(d) $\{(2,2,1),(2,1,2),(2,3,0)\}$

50. Which of the following vectors in $\mathbb{R}^3$ are linearly independent ?

(a) $\{(4,3),(12,9)\}$
(b) $\{(1,2,1),(-1,1,1),(1,-1,-1)\}$
(c) $\{(5,5,5),(2,1,1),(1,1,1)\}$
(d) $\{(8,20),(2,4)\}$